Joanna Piper Morgan, Ph.D.

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EDUCATION

Oregon State University (OSU), Corvallis, OR — September 2020 – June 2025
Ph.D., Mechanical Engineering

- **Minor:** Nuclear Engineering, **Emphasis:** Thermal Fluid Sciences (TFS)
- **Dissertation Title:** Algorithms and Software Engineering Schemes for Time-Dependent Radiation Transport on Heterogeneous Compute Architectures.
- **Academic Advisor:** Kyle E. Niemeyer, **Minor Advisor:** Todd S. Palmer

Oregon State University (OSU), Corvallis, OR — September 2020 – March 2022
M.S., Mechanical Engineering

- **Emphasis:** Thermal Fluid Sciences (TFS)
- **Project Report:** Explorations of Monte Carlo Solution and Implementation Methods for Thermal Radiation and Neutron Transport
- **Academic Advisor:** Kyle E. Niemeyer

Oregon Institute of Technology (OIT), Klamath Falls, OR — September 2016 – June 2020
B.S., Mechanical Engineering, magna cum laude

- **Minors:** Applied Physics; Applied Mathematics

RESEARCH EXPERIENCE

Lawrence Livermore National Lab, Nuclear Criticality Safety Division — September 2025
Postdoctoral Research Scholar
Subject: TEX Experiment design and evaluation, Sn and Monte Carlo algorithms development using MFEM and PETSc.

Oregon State University, School of MIME — June 2020 – June 2025
Graduate Research Assistant
Subject (1): Python based acceleration and abstraction of compute kernels for dynamic Monte Carlo in a rapid methods development code MC/DC as part of the Center for Exascale Monte Carlo Neutron Transport (CEMeNT)
Subject (2): Investigation of the one cell inversion method as an alternative to transport sweeps for deterministic dynamic neutron transport on GPUs
Mentors: Kyle E. Niemeyer & Todd S. Palmer

Advanced Micro Devices (AMD), HPC GPU Apps Support Group — September 2023 – May 2024
Co-Op Research Intern
Subject: Python–Numba compiler bindings for AMD GPUs and Sn numerical methods in rocSOLVER, profiler testing for MI300a
Mentors: Damon McDougall, Christopher Kime

Los Alamos National Laboratory, XCP-3 — June 2022 – May 2023
Graduate Research Intern
Subject: Implementing Woodcock delta tracking on a structured mesh within the production code MCATK. This work was pushed to a production version release.
Mentors: Travis J. Trehan, Timothy P. Burke, & Collin J. Josey

Los Alamos National Laboratory, CCS-2 June 2021 – June 2022
Graduate Research Intern
Subject: Novel methods exploration in vectorizable variance reduction for thermal radiation transport (TRT)
Mentors: Kendra Long & Alex Long

Argonne National Laboratory June 2019 – September 2019
Lee Teng Undergraduate Research Fellow
Subject: Cancer radiotherapy scanner magnet design & analysis
Mentor: Brahim Mustapha

Thomas Jefferson National Accelerator Facility June 2018 – August 2018
Science Undergraduate Laboratory Intern (SULI)
Subject: Superconducting quadrupole magnet optimization
Mentor: Renuka Rajput-Ghoshal

TEACHING EXPERIENCE

Guest Lecture September 2026
Fundamentals of Monte Carlo Neutronics
University California Berkeley, Dept. of Nuclear Engineering, NE156/256M Nuclear Criticality Safety

Webinar September 2026
Rapid Development of a High-Performance Application Using Python **IDEAS Productivity, HPC Best Practices Webinars**

Teaching Assistant December 2026
US Research Software Sustainability Institute (URSSI), Portland, OR

Teaching Assistant January 2024
US Research Software Sustainability Institute (URSSI), Portland, OR

Supplemental Instructor, Peer consultant, Math grader September 2017 – June 2020
Oregon Institute of Technology, Various Depts.

JOURNAL PUBLICATIONS

1. J. P. Morgan, I. Variansyah, K. B. Clements, T. S. Palmer, & Niemeyer, K. E. (2026). Hybrid Delta Tracking Schemes Using a Track-Length Estimator. *Journal of Computational and Theoretical Transport*, 1–33. doi:10.1080/23324309.2026.2618791 arXiv:2510.00152.

2. J. P. Morgan, I. Variansyah, T. S. Palmer, and K. E. Niemeyer. (2025) One-Cell Inversion for Solving Higher-Order Time-Dependent Radiation Transport on GPUs. Accepted. *Nuclear Science and Engineering*. 200(5) 1074-1093 doi:10.1080/00295639.2025.2510004. arXiv:2503.00264.

3. J. P. Morgan, I. Variansyah, B. Cuneo, T. S. Palmer, and K. E. Niemeyer. (2025) Performant and Portable Monte Carlo Neutron Transport via Numba. *Computing in Science and Engineering (IEEE)* 27(1), pp. 57-65. doi:10.1109/MCSE.2025.3550863. arXiv:2409.04668.

4. J. P. Morgan, A. Mote, S. Pasmann, G. Ridley, T. S. Palmer, K. E. Niemeyer, R. G. McClarren. (2024) "The Monte Carlo Computational Summit – October 25 & 26, 2023 – Notre Dame, Indiana, USA". *Journal of Computational and Theoretical Transport*. 53 (5), 361–382. doi:10.21105/joss.06415.

5. J. P. Morgan, I. Variansyah, S. Pasmann, K. B. Clements, B. Cuneo, A. Mote, C. Goodman, C. Shaw, J. Northrop, R. Pankaj, E. Lame, B. Whewell, R. McClarren, T. Palmer, L. Chen, D. Anistratov, C. T. Kelley, C. Palmer, and K. E. Niemeyer. (2024) "Monte Carlo / Dynamic Code (MC/DC): An accelerated Python package for fully transient neutron transport and rapid methods development". *Journal of Open Source Software*. 9(96), 6415. doi:10.1080/23324309.2024.2354401. arXiv:2402.08161.

CONFERENCE PUBLICATIONS

1. J. P. Morgan, O. I. Ortega. Reduced Precision Diffusion Synthetic Acceleration for S_N Neutron Transport in LLNL's ARDRA using Hypre. *Proceedings of the International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (ANS M&C 2026). in review* arXiv:2609.04451

2. J. P. Morgan, J. Glesmann, C. Percher, Z. Lemke, Experiment Completion: Thermal Epithermal eXperiments (TEX) with Iron Diluents for Hanford Tank Safety Basis. (2026) *Transactions of the American Nuclear Society*, Phoenix, AZ.

3. J. P. Morgan. Second Moment Acceleration for Incident Fluxes on Cell Surfaces. (2026) *29th International Conference on Transport Theory (ICTT-29).*

4. J. P. Morgan, C. Percher, E. Aboud. Revision Two and Three Comparison from Flattop HEU Benchmark (HEU-MET-FAST-028). (2026) *Transactions of the American Nuclear Society*, volume 134. pp 407-410 Denver, CO. doi:10.13182/T134-11188

5. J. P. Morgan, B. Cuneo, I. Variansyah, K. E. Niemeyer. (2025) "Enabling GPU portability into the Numba-JITed Monte Carlo particle transport code MC/DC". *Proceedings of the International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (ANS M&C 2025).* Denver, CO, USA. doi:10.13182/MC25-47142. arXiv:2501.05440.

6. B Cuneo, J. P. Morgan, I. Variansyah, K. E. Niemeyer. (2025) "Comparing the Performance of MC/DC's on-GPU Event-based Processing Methods in Multigroup and Continuous-energy Problems". *Proceedings of the International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (ANS M&C 2025).* Denver, CO, USA. doi:10.13182/MC25-47174. arXiv:2506.00263.

7. J. P. Morgan, I. Variansyah, T. S. Palmer, and K. E. Niemeyer. (2023) "Exploring One-Cell Inversion Method for Transient Transport on GPU." *Proceedings of the International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering.* Niagara Falls, Ontario, Canada. arXiv:2305.13555.

8. J. P. Morgan, T. J. Trahan, T. P. Burke, C. J. Josey, and K. E. Niemeyer. (2023) "Hybrid-Delta Tracking on a Structured Mesh in MCATK." *Proceedings of the International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering.* Niagara Falls, Ontario, Canada. arXiv:2306.07847.

9. I. Variansyah, J. P. Morgan, K. E. Niemeyer, and R. G. McClarren. (2023) "Development of MC/DC: a performant, scalable, and portable Python-based Monte Carlo neutron transport code." *Proceedings International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering.* Niagara Falls, Ontario, Canada. arXiv:2305.07636.

10. J. P. Morgan, A. Long, K. Long, and K. E. Niemeyer. (2022) "Novel MC TRT Method: Vectorizable Variance Reduction for Energy Spectra" In *Transactions of the American Nuclear Society*, volume 126, pp. 276–278. Anaheim, California, USA. doi:10.13182/T126-38137. zenodo:6646813.

11. J. P. Morgan, T. S. Palmer, and K. E. Niemeyer. (2022) "Explorations of Python-Based Automatic Hardware Code Generation for Neutron Transport Applications." In *Transactions of the American Nuclear Society*, volume 126, pp. 318–320. Anaheim, California, USA. doi:10.13182/T126-38066. zenodo:6643659.

PUBLICATIONS IN PREPARATION AND REVIEW

1. J. P. Morgan, J. Norris. C. Percher, D Heinrichs. Sensitivity of Criticality Benchmarks to Unresolved-Resonance Probability Tables. *in review* arXiv:2609.04451

2. J. P. Morgan, B. S. Spithworth H. Aldani. Application of a GenEO RAS preconditioner to Sn radiation transport on DG meshes. *in review.* arXiv:2609.04451

3. J. P. Morgan. Unnamed paper on Inconsistent second moment itterative methods for one-cell Inversions. *in review.*

4. J. P. Morgan, T. S. Palmer, and K. E. Niemeyer. Boundary Projection Acceleration for 2D one-cell Inversions in the Scattering Limit. *in preparation* arXiv:2609.04451

5. M. Derman, J. P. Morgan, K. E. Niemeyer, T. S. Palmer. Unnamed paper on complex eigenvalue decomposition. *in preparation*

CONFERENCE AND POSTER PRESENTATIONS

- J. P. Morgan. Experiment Completion: Thermal Epithermal eXperiments (TEX) with Iron Diluents for Hanford Tank Farm. Winter Meeting of the American Nuclear Society. November 2026.

- J. P. Morgan. Second Moment Acceleration for Incident Fluxes on Cell Surfaces. 29th International Conference on Transport Methods (ICTT-29). June 2026 poster

- J. P. Morgan, C. Percher, E. Aboud. Revision Two and Three Comparison from Flattop HEU Benchmark (HEU-MET-FAST-028). (2026) Annual Meeting of the American Nuclear Society May 2026

- Jesse Norris, J Morgan, I Snider, C Percher, D Heinrichs, S Kim. LLNL Testing of ENDF/B-VIII.1 with the COG Validation Suite. Cross Section Evaluation Working Group (CSEWG). January 2026 slides

- J. P. Morgan, B. Cuneo, I. Variansyah, K. E. Niemeyer. May 2025. Enabling GPU Portability into the Numba-JITed Monte Carlo Particle Transport Code MC/DC. International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (M&C 2025), Denver, CO

- J. P. Morgan. February 2025. AMD Tools Development Workshop, University of Oregon, Eugene, OR

- J. P. Morgan, K. E. Niemeyer. July 2024. Performant and Portable High-Performance Computing at Scale via Python and Numba. SciPy. Tacoma, WA

- J. P. Morgan. "Exploring One-Cell Inversions for Transient Transport on GPUs." Copper Mountain Conference on Iterative Methods (SIAM). April 2024.
- J. P. Morgan. "High Performance Python for Rapid Methods Development in Monte Carlo / Dynamic Code." Sustainable Scientific Software Conference. April 2024.
- J. P. Morgan, I. Variansyah, T. S. Palmer, and K. E. Niemeyer. "Exploring One-Cell Inversion Method for Transient Transport on GPU." High Energy Density Summer School, San Diego, California. July 2023
- J. P. Morgan, I. Variansyah, T. S. Palmer, and K. E. Niemeyer. "Exploring One-Cell Inversion Method for Transient Transport on GPU." International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering. Niagara Falls, Ontario, Canada (2023).
- J. P. Morgan, T. J. Trahan, T. P. Burke, C. J. Josey, and K. E. Niemeyer. "Hybrid-Delta Tracking on a Structured Mesh in MCATK." International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering. Niagara Falls, Ontario, Canada (2023).
- J. P. Morgan, T. S. Palmer, & K. E. Niemeyer, "Hardware Code Generation Techniques for Accelerating Python.", Scientific Python, Austin, Texas, USA, July 2022
- J. P. Morgan, T. S. Palmer, & K. E. Niemeyer, "Hardware Code Generation Techniques for Accelerating Python", Annual Meeting of the American Nuclear Society, Anaheim, California, USA, June 2022
- J. P. Morgan, A. Long, K. Long & K. E. Niemeyer, "A novel MC TRT method: vectorizable variance reduction for the energy spectra", Annual Meeting of the American Nuclear Society, Anaheim, California, USA, June 2022
- J. P. Morgan and B. Mustapha, "Carbon therapy X-Y scanner magnet analysis," in Lee Teng Internship - Posters and Final Report Presentations, Fermi National Accelerator Facility, Batavia, Illinois. August 2019.
- J. P. Morgan and R. Rajput-Ghoshal, "Jefferson lab electron ion collider interaction region quadrupole magnet optimization," in Undergraduate Research Poster Presentations, Thomas Jefferson National Accelerator Facility, Newport News, Virginia, August, 2018.

CONFERENCE AND TECHNICAL MEETING ATTENDANCE

** presented at*

- September 2027, 13th International Conference on Nuclear Criticality (ICNC), Manchester, United Kingdom*
- April 2027, International Criticality Safety Benchmark Evaluation Project (ICSBEP), Technical Review Meeting, Japan*
- April 2027, International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering 2027 (M&C 2027), Columbus, OH*
- November 2026, 2026 ANS Winter Confrence & Expo, Phoenix, AZ*

- June 2026, 29th International Conference on Transport Theory (ICTT-29)*, Aix-en-Provence, France
- June 2026, American Nuclear Society Annual Conference*, Denver, CO
- April 2026, International Criticality Safety Benchmark Evaluation Project (ICSBEP) Technical Review Meeting, Paris, France
- January 2026, Nuclear Data Week (Cross Section Evaluation Working Group (CSEWG)*, Brookhaven National Lab, NY
- May 2025, International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (M&C 2025), Denver, CO*
- July 2024, Scientific Computing in Python (SciPy), Tacoma, WA*
- May 2024, NEA Workshop for Radiation Transport Simulation Developers (RTS 2024), Frascati, Italy
- April 2024, 18th Copper Mountain Conference on Iterative Methods (SIAM), Copper Mountain, CO*
- April 2024, Sustainable Scientific Software Conference (S3C), Seattle, WA*
- January 2024, NUWEST, Albuquerque, NM (unable to attend due to weather)
- August 2023, International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (M&C 2023), Niagara Falls, Ontario, Canada
- July 2022, Scientific Computing in Python (SciPy), Austin, TX*
- June 2022, American Nuclear Society Annual Conference, Anaheim, CA*
- October 2021, Conference on Mathematics and Computational Methods Applied to Nuclear Science (ANS M&C 2021), Raleigh, North Carolina (virtual)

PROFESSIONAL DEVELOPMENT

Criticality Safety Engineer Two Week Hands-On Trainings and Education Course Aug 3–14, 2026
US Dept. of Energy, Nuclear Criticality Safety Program

AMD Tools Development Workshop February 2025
University of Oregon, Eugene, OR

High Energy Density Science (HEDS) Summer School July 2023
University of California San Diego, San Diego, CA

Fundamentals of Accelerator Physics and Technology with Simulations and Measurements Lab June 2018
2018 United States Particle Accelerator School (USPAS) Albuquerque, NM

PROFESSIONAL AFFILIATIONS

- Member, American Nuclear Society (ANS)
- Member, Out in Science Technology Engineering and Mathematics (oSTEM)
- Member, Tau Beta Pi (Engineering Honors Society), Oregon Delta 2018

OTHER ACTIVITIES

- Volunteer; Whiteside Theater, Corvallis, OR (March 2023 – June 2025)
- Student Commissioner; Library Recourse Commission (October 2017 – June 2020)
- Committee Member; University Librarian Search Committee (March 2018 – June 2018)
- Student Building Manager; Oregon Institute of Technology College Union (February 2017 – March 2020)
- President; Oregon Institute of Technology Chapter of Circle K International Community Service Club (May 2019 – June 2020)
- Treasurer; Oregon Delta – Tau Beta Pi (May 2019 – May 2020)
- Notary Public; State of Oregon (November 17th, 2016 – November 16th, 2020)

AWARDS AND FELLOWSHIPS

- Illinois Accelerator Institute; Lee Teng Undergraduate Research Fellowship in Accelerator Science (2019)
- ASME; Irma and Robert Bennett Scholarship (2019)
- Pride Foundation; Lenehan-Warn Technical Education Scholarship (2017)

SKILLS

- **Clearance:** Active Q security clearance
- **Languages:** PETSc (beginner-intermediate), MFEM (beginner-intermediate), C/C++ (OpenMP, C-CUDA, HIPC C/ROCm, Kokkos, AVX / SIMD, MPI, Intel MKL, LAPACK) (beginner-intermediate), Python (Numba, mpi4py, Numpy, Scipy, Matplotlib, CuPy, CProfiler) (intermediate), Matlab (intermediate), VBA in Excel, FORTRAN (beginner), LLVM (beginner-intermediate)
- **Compilers:** Numba (LLVM bindings for Python), Intel Compilers, HIP and ROCm (Clang) compilers, GCC, CRAY Compilers
- **Software Development Tools:** Profilers (Intel Vtune, NVIDIA N-Sight, TotalView), Debuggers (valgrind, gdb)
- **Engineering Design and Simulation Tools:** CAD/CAE (Creo, SolidWorks, AutoCAD, Inventor, NX, CST Studio Suite (E&M Studio), Opera, FEMM)
- **Nuclear Engineering Codes:** MCNP, PartiSn, SCALE6.2 (Shift (developed in), KENO, Denovo, Tsunami, Origin, XS-DRIN), MCATK (developed in), MC/DC (developed in), COG, NJOY, PrePro, OpenMC
- **Visualization Toolkits:** Paraview, Visit
- **MISC Development Tools:** Services (Git(hub), Gitlab, Bitbucket), Terminal (Bash, z-shell, ssh & X-11 forwarding), Package Publishing (Conda, PyPi, Sphinx, Read the Docs), Continuous Integration (Github actions, Jakamar runners), Website Builders (Sphinx, Read the Docs, Jekyll, Ruby), LaTeX

REFERENCES

Available upon request